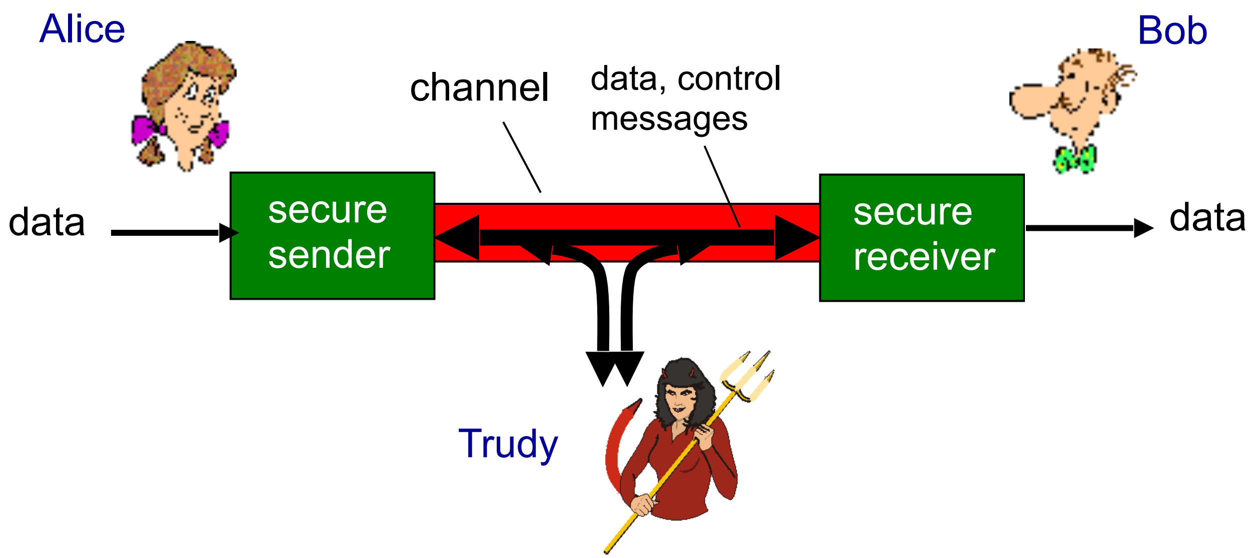
CS151: Encryption and blockchain

Speaker: Xiaoxue (Eira) Zhang

Friends and enemies: Alice, Bob, Trudy

- well-known in network security world
- Bob, Alice (lovers!) want to communicate "securely"
- Trudy (intruder) may intercept, delete, add messages



Who might Bob, Alice be?

- … well, real-life Bobs and Alices!
- Web browser/server for electronic transactions (e.g., on-line purchases)
- on-line banking client/server
- DNS servers
- other examples?

There are bad guys (and girls) out there!

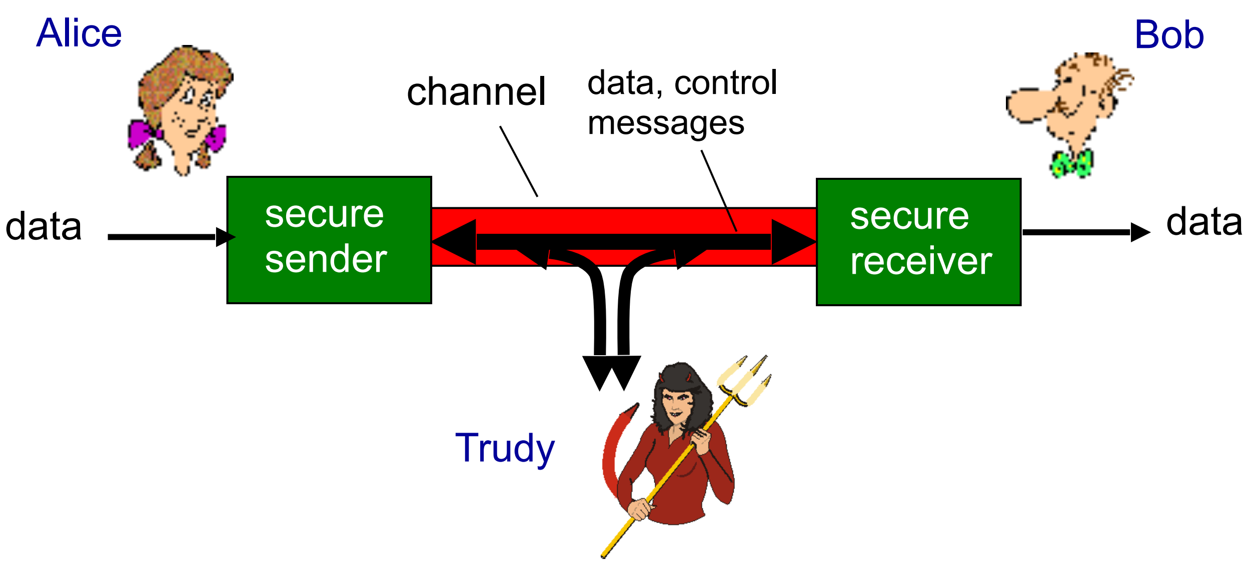
Q: What can a "bad guy" do?

A: A lot!

- eavesdrop: intercept messages
- actively insert messages into connection
- impersonation: can fake (spoof) source address in packet (or any field in packet)
- hijacking: "take over" ongoing connection by removing sender or receiver, inserting himself in place
- denial of service: prevent service from being used by others (e.g., by overloading resources)

Secure communication

- Confidentiality: Keep data and communication secret
  - Encryption / decryption
- Authenticity: Parties cannot be impersonated
  - "Did Alice really send this message?"
- Integrity: Messages cannot be modified
  - "Was this the original message that was sent?"

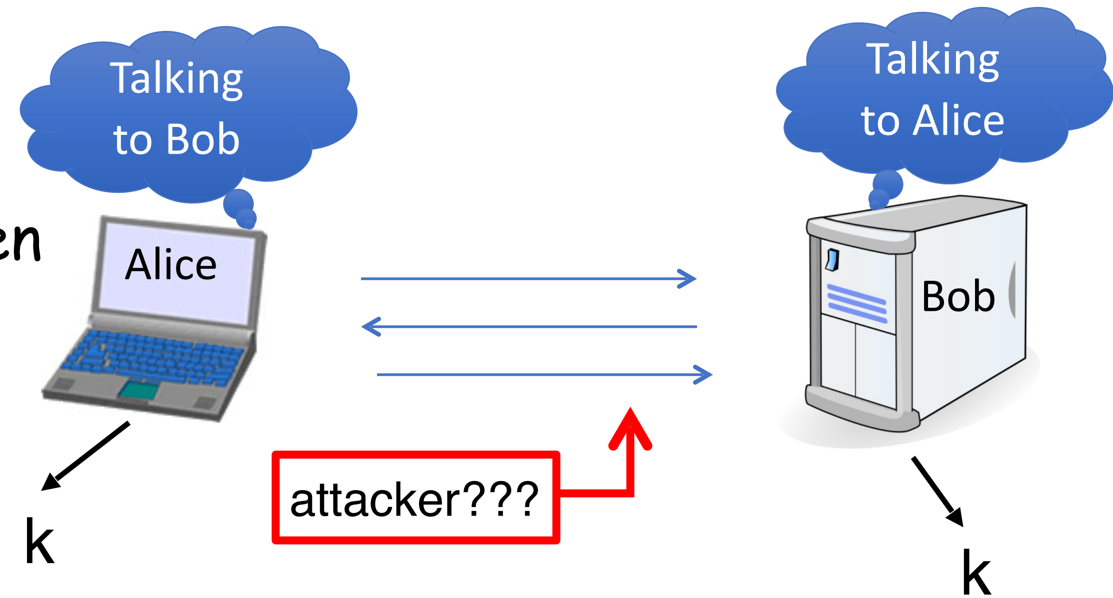


Introduction

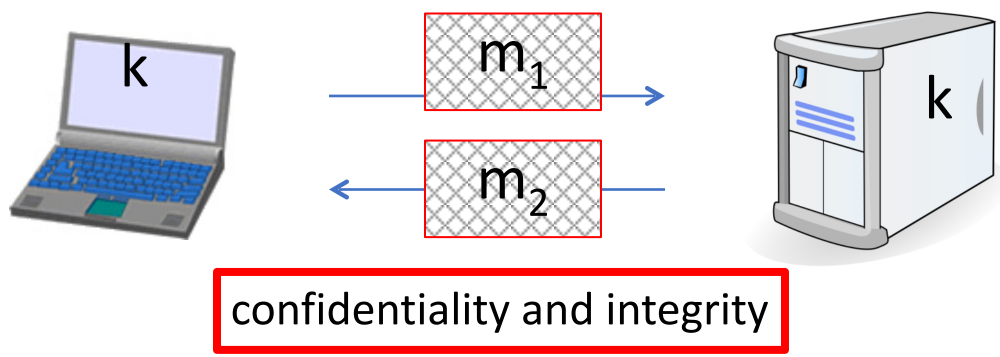
What is cryptography?

Core of Cryptography

Secret key establishmen



Secure communication:

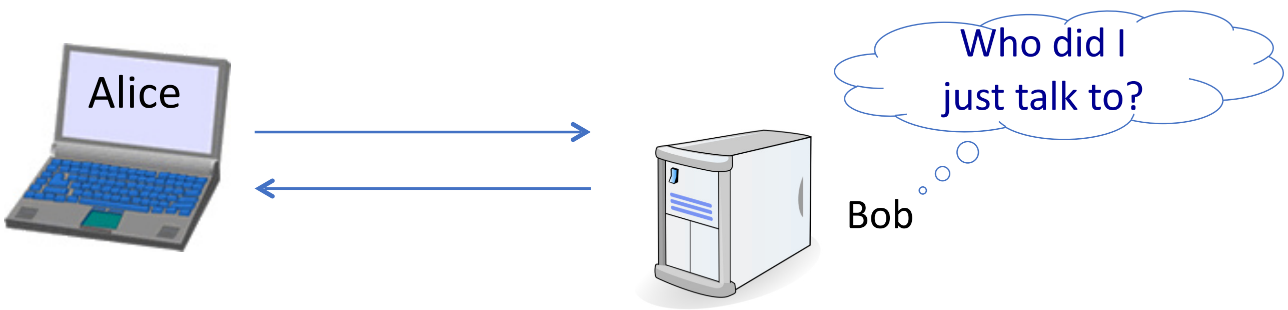


But crypto can do much more

- Digital signatures
  - Sender (Bob) digitally signs document, establishing he is the document owner/creator
  - Make the digital signature via function of the content being signed
  - Verifiable, non-forgeable
    - recipient (Alice) can prove to someone that Bob, and no one else (including Alice), has signed document

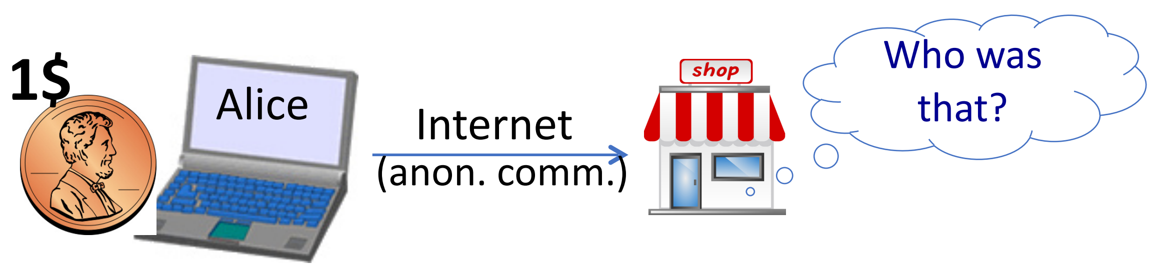
But crypto can do much more

- Anonymous communication
  - Mix network: Alice sends messages to Bob through a sequence of proxies
  - Messages are encrypted and decrypted appropriately
  - Bob doesn't know who is the sender



But crypto can do much more

- Anonymous **digital** cash
  - Can I spend a "digital coin" without anyone knowing who I am?
  - How to prevent double spending?

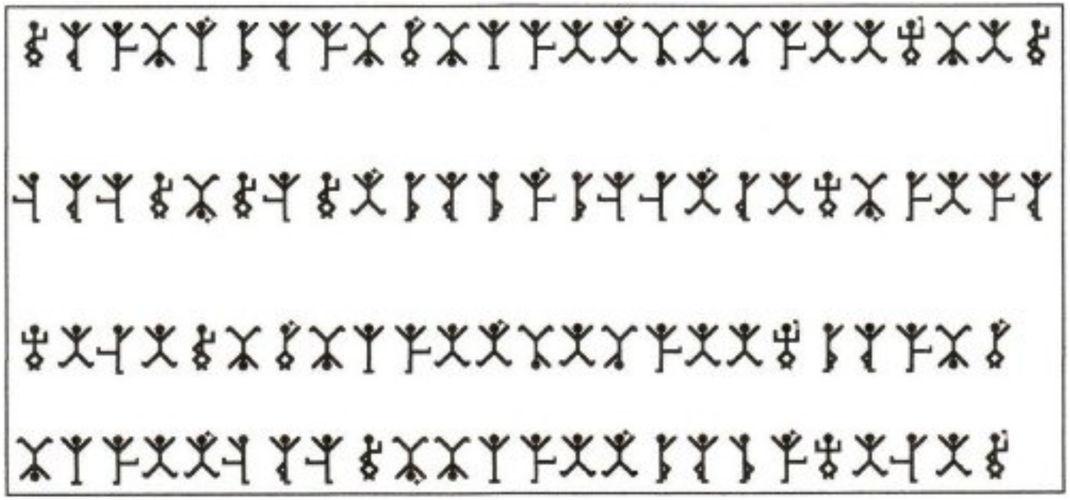


Introduction

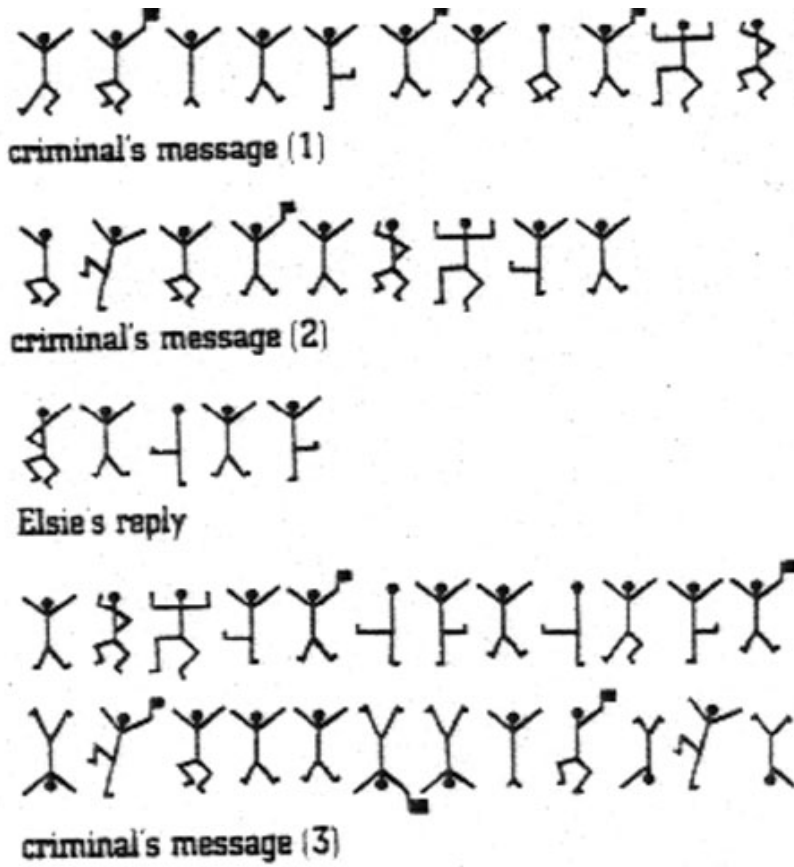
History

The dancing men

- A story about Sherlock Holmes, by Sir Arthur Conan Doyle.



The dancing men



Few Historic Examples (all badly broken)

Substitution cipher: substituting one thing for another

$$c := E(k, \text{"}bcza\text{"}) = \text{"}wnac\text{"}$$

$$\mathrm{D}(k, c) = \text{"}bcza\text{"}$$

$$k := \boxed{\begin{array}{l} \mathrm{a} \to c \\ b \to w \\ \mathrm{c} \to n \\ \vdots \\ \mathrm{z} \to a \end{array}}$$

Key: substitution table

How to break a substitution cipher?

(1) Use frequency of English letters

"e": 12.7%, "t":9.1%, "a": 8.1%

(2) Use frequency of pairs of letters (digrams)

common pairs: "he", "an", "in", "th"

ciphertext-only attack

An Example

UKBYBIPOUZBCUFEEBORUKBYBHOBBRFESPVKBWFOFERVNBCVBZPRUBOFE
RVNBCVBPCYYFVUFOFEIKNWFRFIKJNUPWRFIPOUNVNIPUBRNCUKBEFWWFD
NCHXCYBOHOPYXPUBNCUBOYNRVNIWNCPOJIOFHOPZRVFZIXUBORJRUBZR
BCHNCBBONCHRJZSFWNVRJRUBZRPCYZPUKBZPUNVPWPCYVFZIXUPUNFCP
WRVNBCVBRPYYNUNFCPWWJUKBYBIPOUZBCUIPOUNVNIPUBRNCHOPYXPUB
NCUBOYNRVNIWNCPOJIOFHOPZRNCRVNBCUNENVVFZIXUNCHPCYVFZIXUPU
NFCPWZPUKBZPUNVR

B	**36**
N	**34**
U	**33**
P	**32**
C	**26**

→ E

→ T

→ A

NC	**11**
PU	**10**
UB	**10**
UN	**9**

digrams

→ IN

→ AT

UKB	**6**
RVN	**6**
FZI	**4**

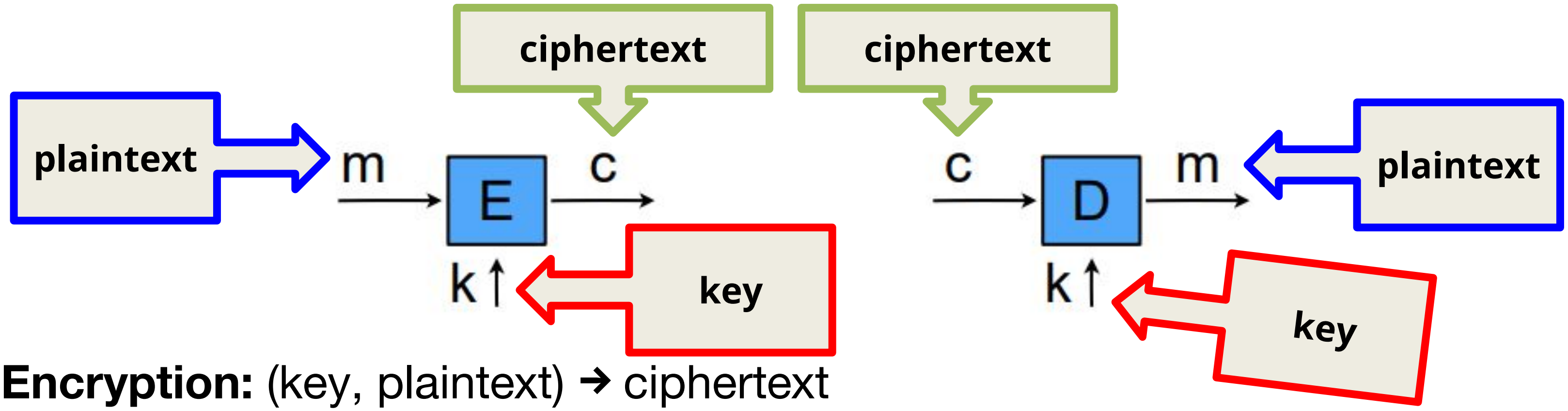
trigrams

→ THE

Symmetric-key encryption

- **Encryption:** (key, plaintext) ➔ ciphertext
  - ➢ $E_k(m) = c$
- **Decryption:** (key, ciphertext) ➔ plaintext
  - ➢ $D_k(c) = m$
- **Functional property:** Where $D_k(E_k(m)) = m$

Symmetric-key encryption



- **Encryption:** (key, plaintext) → ciphertext
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- **Decryption:** (key, ciphertext) → plaintext
  - ➢ $D_k(c) = m$
- **Functional property:** Where $D_k(E_k(m)) = m$

The One Time Pad (Vernam 1917)

$$c := E(k, m) = k \oplus m$$
$$m := D(k, c) = k \oplus c$$

msg:	0	1	1	0	1	1	1	$\oplus$
key:	1	0	1	1	0	1	0	
CT:	1	1	0	1	1	0	1	

$$D(k, E(k, m)) = D(k, k \oplus m) = k \bigoplus \left(k \bigoplus m\right) = m$$

The One Time Pad (Vernam 1917)

You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?

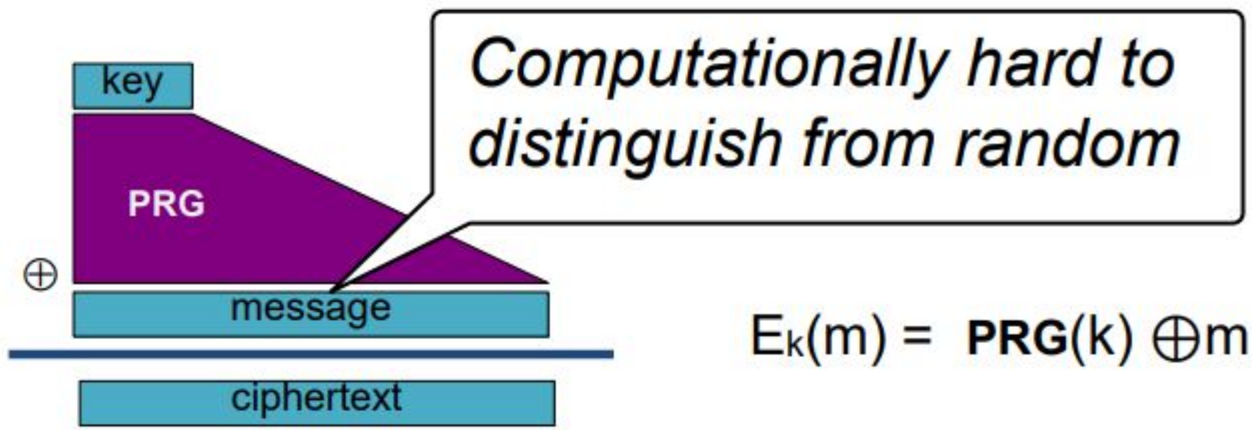
Yes, the key is $k = m \oplus c$.

The One Time Pad (Vernam 1917)

- Shannon (1949)
  - ➢ Information-theoretic security: without key, ciphertext reveals no "information" about plaintext
- Problems with OTP
  - Can only use key once
  - Key is as long as the message
  - No integrity protection

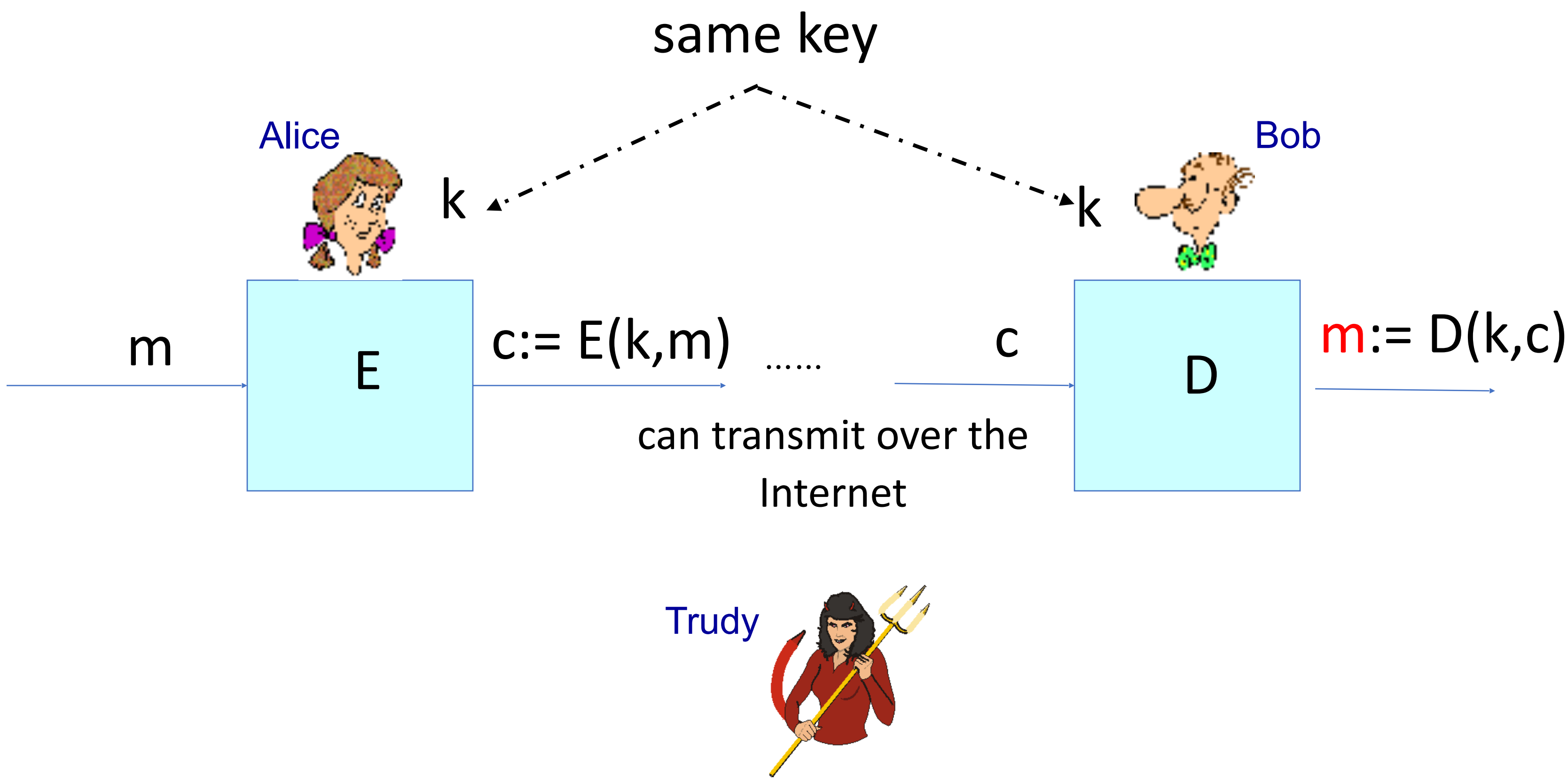
Stream Ciphers: making OTP practical

- Problem: OTP key is as long as message
- Solution: Pseudo random generator
- Stream ciphers uses bit-by-bit to generate ciphertext.

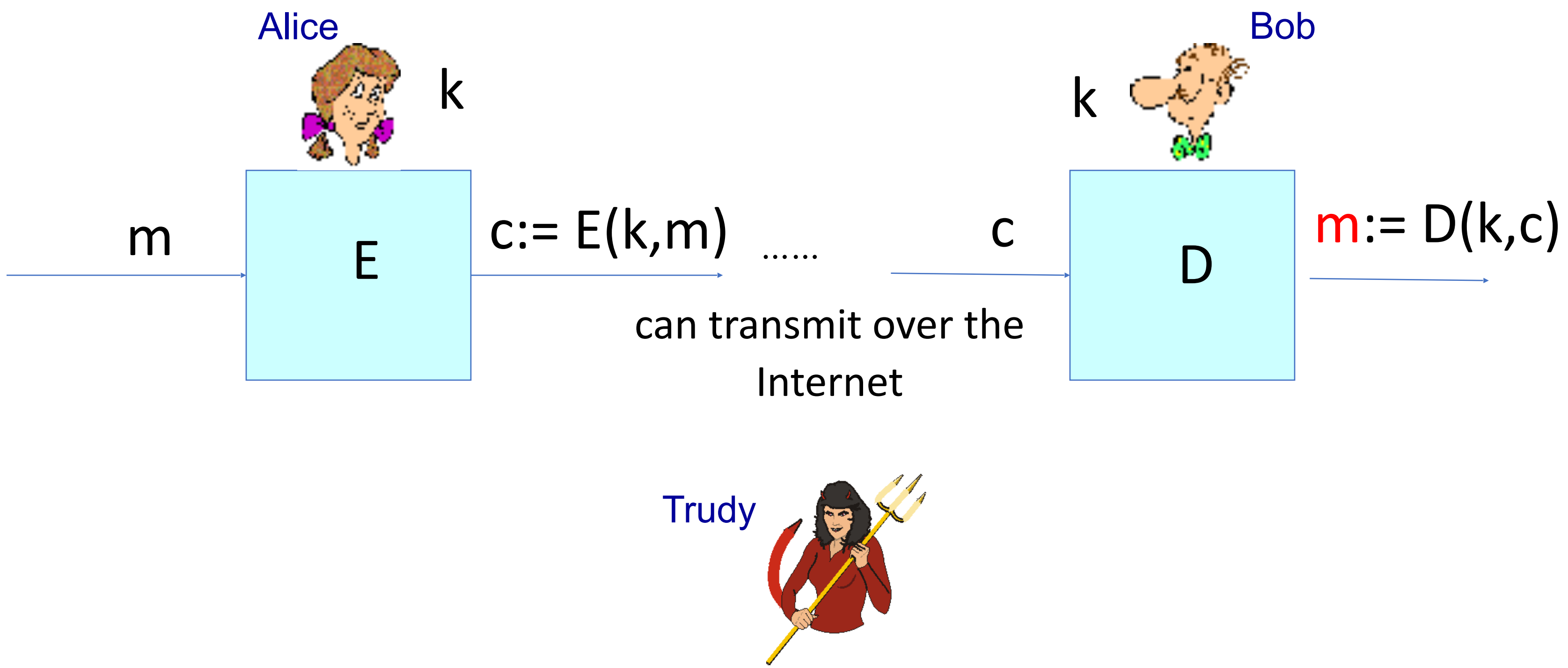


Examples: ChaCha, Salsa, etc.

Symmetric Ciphers



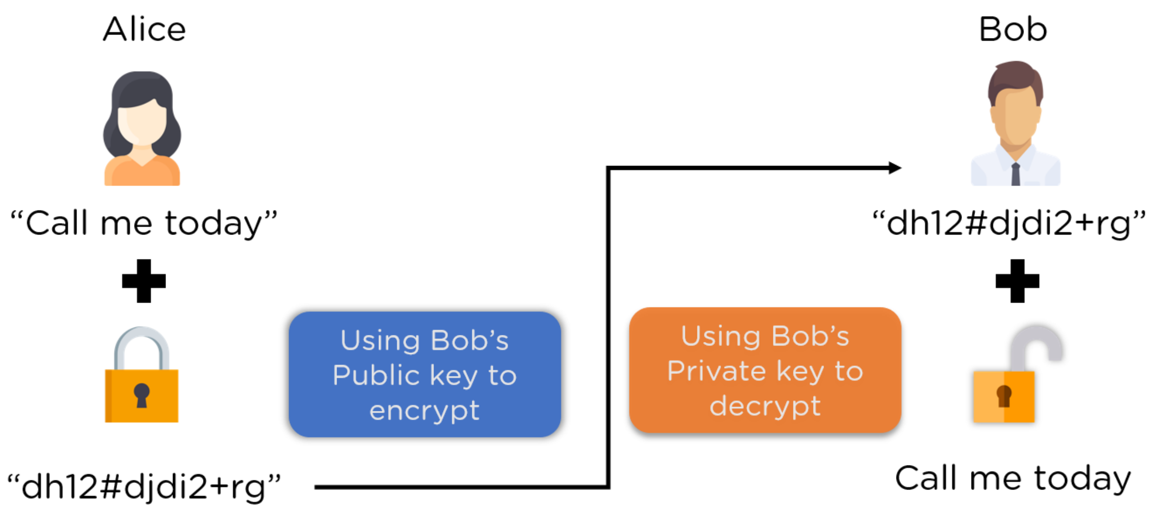
Main challenge:
How to share K safely to begin with?



Asymmetric encryption

Use two different keys:

- Public Key – shared with everyone
- Private Key – kept secret by the owner

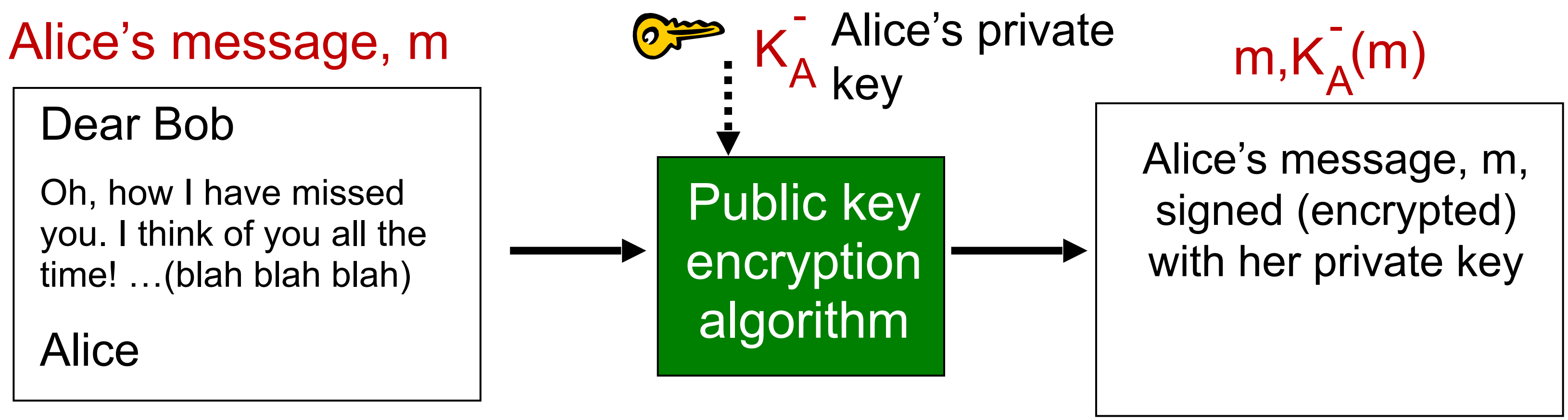


Asymmetric encryption

Authentication

simple digital signature for message m:

- Alice signs m by encrypting with her private key K_A^-, creating "signed" message, $K_A^-(m)$



From Encryption to Blockchain

Cryptography gave us the tools to:

- Keep data secret (encryption)
- Verify identity (digital signatures)
- Detect tampering (hashing)

Blockchains combine these tools to solve a new problem:

How do we keep a shared database secure when no one is fully trusted?

Six reasons to care about Blockchains

Reason 1: You like excitement!

10,000 BTC
= 2 pizzas

Source: CNBC

45 BTC = Lambo

Reason 1: You like excitement!

Reason 2: You like depression

Reason 3: You like rollercoasters

Reason 4: You have a sense of the absurd

Reason 5: Everyone else cares

Forbes

Insurance Interrupted: How Blockchain Innovation is Transforming the Insurance Industry

Andrea Tinianow Contributor
Crypto & Blockchain
I am the Blockchain Czarina. I bring

Banking and Fintech on The Blockchain

Gerald Fenech Contributor
Crypto & Blockchain
I am a journalist writing

Financial Ledger

3,686 views | Dec 19, 2018, 07:10am

Why The Travel Industry Blockchain's Best Bet

David Petersson Contributor
Crypto & Blockchain

Forbes CommunityVoice Connecting expert communities to the Forbes audience. What is This?

3,202 views | Mar 12, 2018, 09:00am

In 2018, Blockchain Will Usher In A New Era Of Proptech

Joshua Fraser CommunityVoice
Forbes Real Estate Council CommunityVoice
Real Estate

POST WRITTEN BY

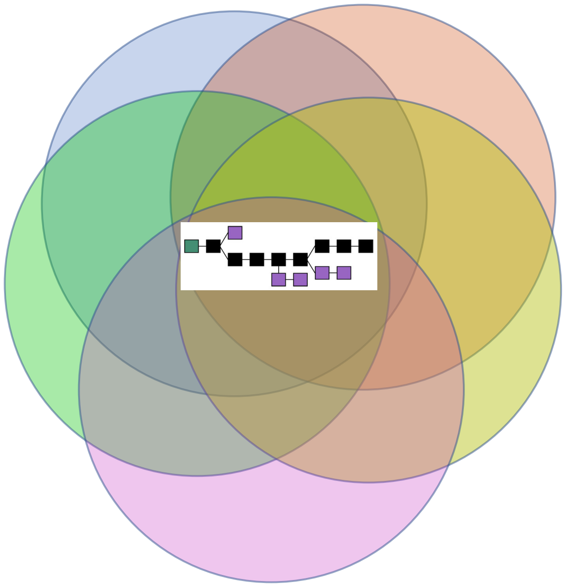
Joshua Fraser

CEO at Data Nerds and Estated. Focused on bringing transparency into residential/commercial real estate through data and analytics.

Reason 6: They are a fascinating technology spanning many areas of computer science–and beyond

For example:

- Distributed systems
- Programming languages
- Formal verification
- Cryptography / security
- Game theory
- Technology law
- Business



Background

- Centralized trading system

Background

- Centralized DNS

A trusted third party

Background

- Centralized PKI

A trusted third party

Background

- Problem
  - Single point of failure

- What if the system is decentralized and there is no trusted third party?
  - We need a distributed ledger

(blockchain)

What is a blockchain?

- It looks like a file
- Append-only global log
- Every node on the network has a consistent copy

What is a blockchain?

- A distributed system of computers…
  - running on a peer-to-peer network…
  - with the ability to run arbitrary user programs (smart contracts)…
  - that can program money/assets.

🏆 ***Anyone*** can join the protocol (Permissionless)

🏆 ***Not*** owned or controlled by any company or country. (Decentralized)

🏆 Very strong security, reliability, and neutrality.

🏆🏆🏆 A **trustworthy "world computer" for everyone.**

Some blockchain use cases

Platform for digital assets (***cryptocurrency***) with growing adoption

- 40% of American adults now own "crypto" (report by security.org)
- Bitcoin and Ethereum ETF (Exchange-Traded Funds)

Statement on the Approval of Spot Bitcoin Exchange-Traded Products

Chair Gary Gensler

Jan. 10, 2024

U.S. Securities and Exchange Commission

Will Schmitt in New York JULY 22 2024 FINANCIAL TIMES

SEC approves ether ETFs as crypto moves closer to mainstream

Latest authorisations follow January launches of first US spot bitcoin ETFs

Some blockchain use cases

Global aid & remittance payments:

- $225M in cryptocurrency donations sent to Ukraine
- OFAC-authorized humanitarian aid to 60,000 healthcare workers in Venezuela, using USDC and VPNs evaded interception by the local regime

Ukraine Bought Weapons, Drones With Crypto Donations

The wartorn nation also revealed it bought non-lethal equipment in a new report detailing its expenditures from crypto donations.

By Amitoj Singh   Aug 17, 2022 at 12:17 p.m. EDT   Upda

Circle Partners with Bolivarian Republic of Venezuela and Airtm to Deliver Aid to Venezuelans Using USDC

https://www.circle.com/blog/circle-partners-with-bolivarian-republic-of-venezuela-and-airtm-to-deliver-aid-to-venezuelans-using-usdc

Some blockchain use cases

DeFi (Decentralized Finance): Financial systems run on "blockchain computer"

- Users *don't* need to trust companies for their funds
- Auditing can be facilitated by *formal verification* of smart contract code
- Open interoperability & innovation

FTX

"with some simple tweaks to computer code," [SBF] helped Alameda Research **misappropriate as much as $65 billion** from FTX customers.

Smart-contract-powered Decentralized Exchanges won't have the same vulnerability!

Some blockchain use cases

A more human passport for the Internet. More powerful. More integrations.

- Platform for data recording/tracking
  - Identity/Naming systems
  - Access-control systems
  - …

Technology Reuters

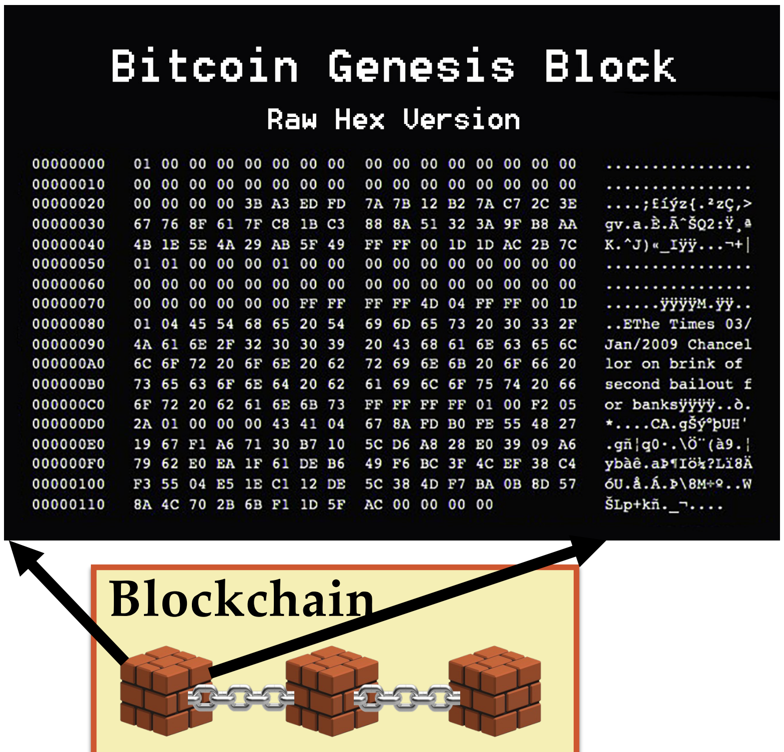
California DMV puts 42 million on blockchain to fight fraud

By **Akash Sriram**

July 30, 2024 6:59 PM EDT · Updated a month ago

```
Handshake is a decentralized, permissionless
naming protocol where every peer is
validating and in charge of managing the root
DNS naming zone with the goal of creating an
alternative to existing Certificate
Authorities and naming systems. Names on the
```

Blockchain's Origin

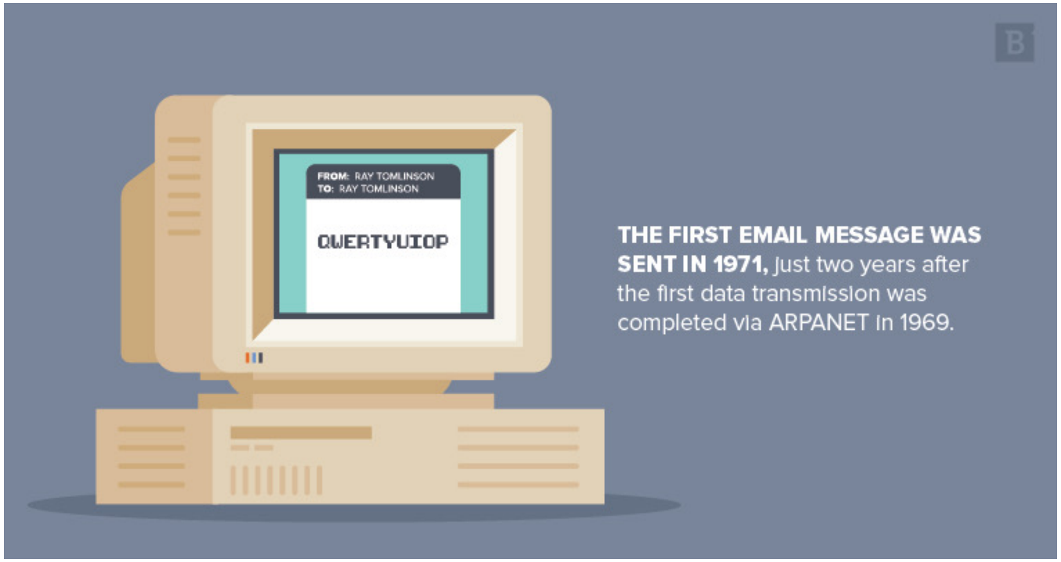


Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

Time travel to 1971



Fun fact: Email systems are early examples of decentralized systems.

- You'd think "hmm, it would be great if I can send $$$ over e-mail"
- How would you do it?

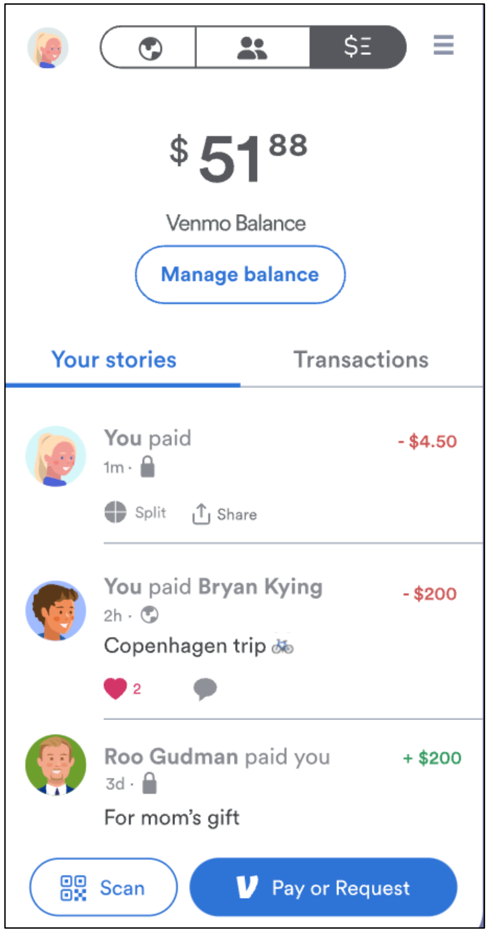
First attempt

- Take a picture of your banknote & send money.jpeg as email attachment?

Seems like we need a central party

- Assume a central party accessible by all
- This party is a Trusted Third Party (TTP)
- Rough idea (v1)
  - User sets up account with TTP and deposits $$$
    - Depositing can be done by mailing a check
  - For Alice to pay Bob, she sends an email to TTP, CCing Bob. TTP moves numbers on its balance book.
  - Withdraw: TTP sends a check.
- Double spending solved!
  - Because TTP is involved in all transactions.

We successfully enabled sending money over emails!



- We are all familiar with the modern version of this idea: PayPal, Venmo, WeChat Pay, Zelle, etc
- With much improved user experiences
  - E.g., Email interaction replaced w/ HTTP
- But the core idea is the same…

Pros / Cons of using TTP (v1)

Pros:

- Dirt simple!
- Fast transactions
- TTP can fix errors / reverse transactions

Cons:

- No privacy from TTP
- TTP can cheat users

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto

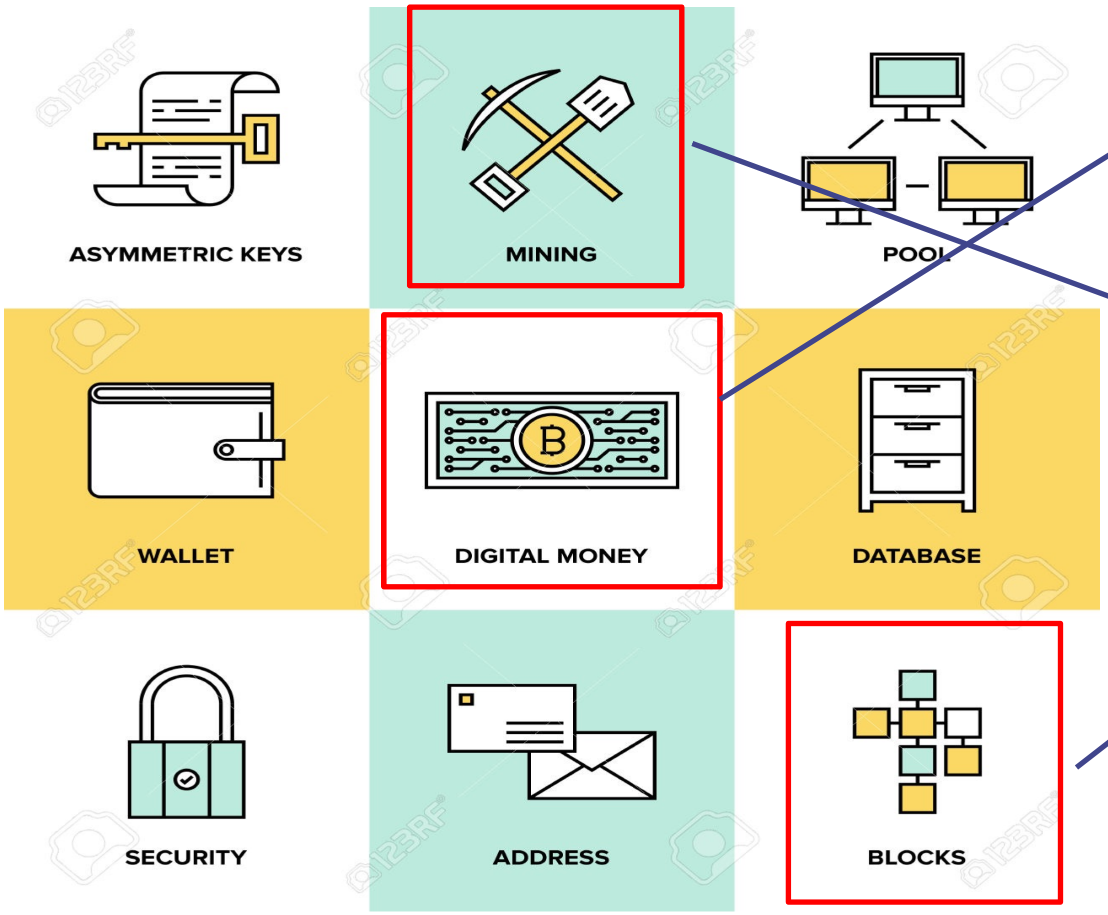
[1] Nakamoto, Satoshi. "Bitcoin: A peer-to-peer electronic cash system." (2008): 28.

Motivation

- Design a decentralized system to conduct direct transaction between any two parties
- Solve the double-spending problem
- Rely on proof instead of trust
  - Everyone can’t be trusted in the network

BLOCKCHAIN

Bitcoin components



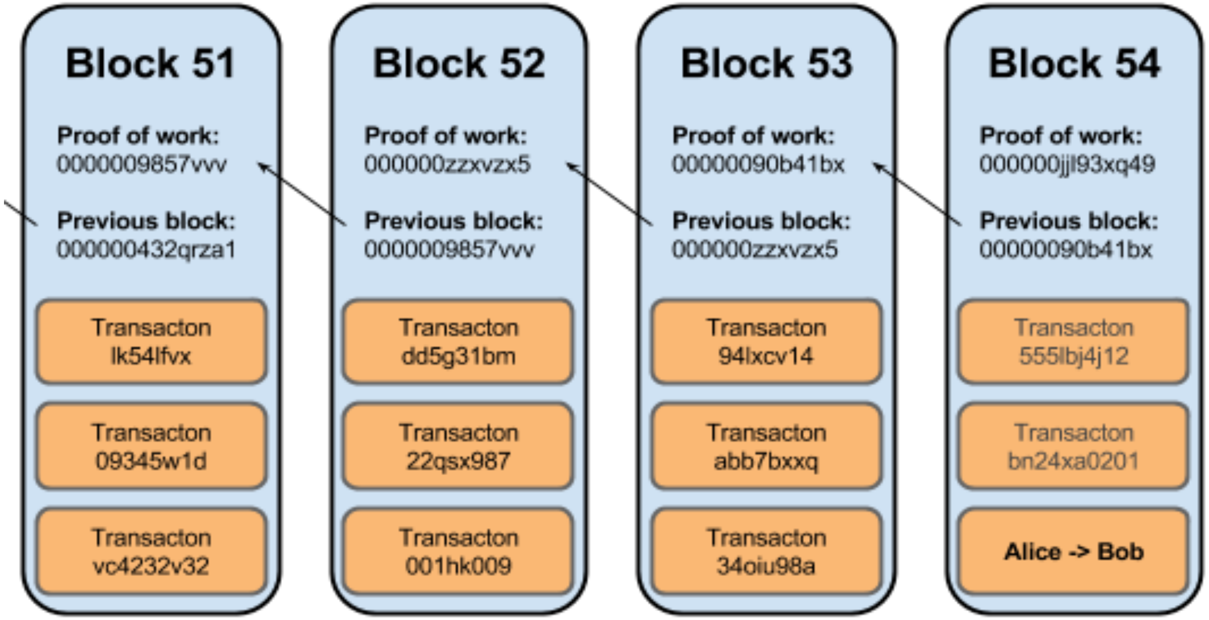
A list of transactions

Miners that mine the block in the network

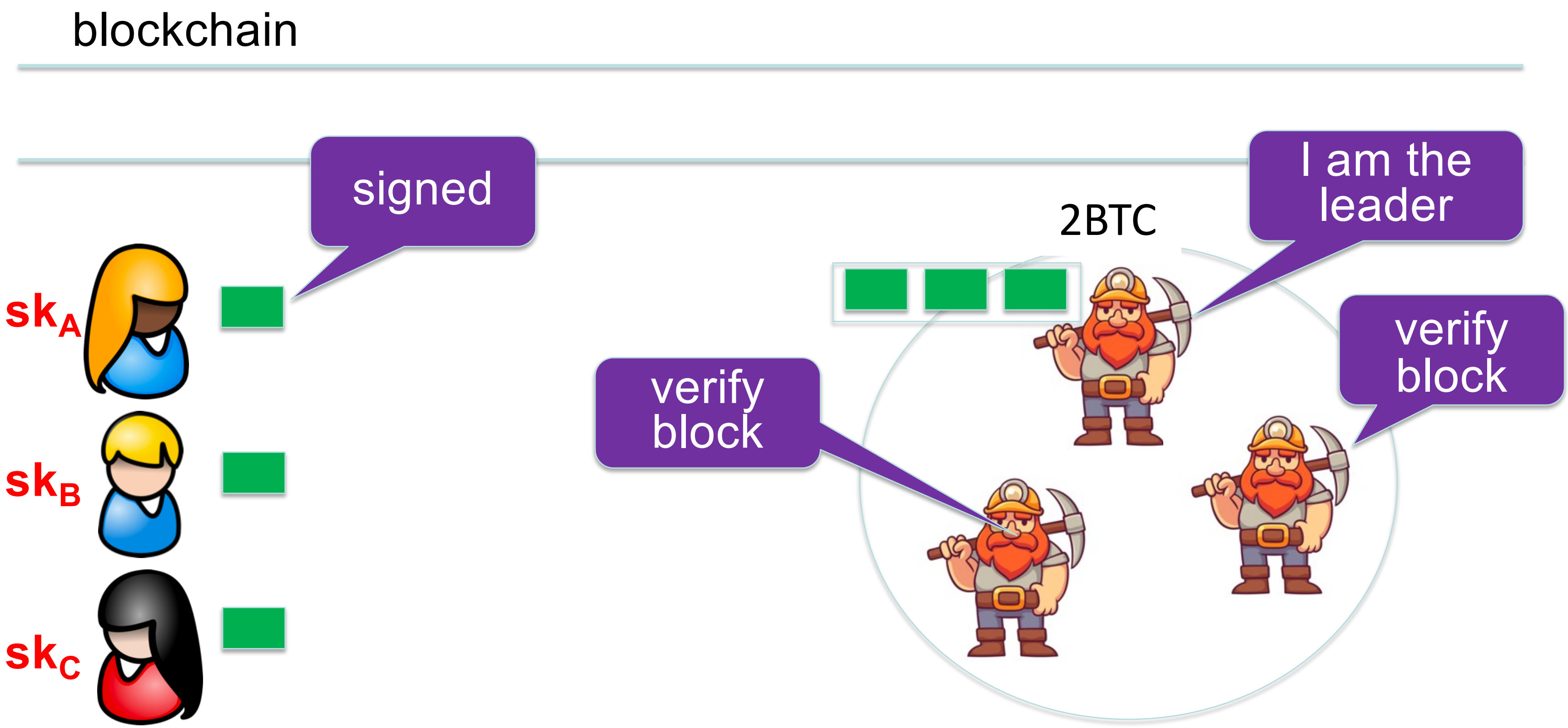
Blockchain that records the valid transactions

Blockchain

- Transactions are organized as blocks
- Blocks are linked
- Verify every transaction when adding a transaction into a block.



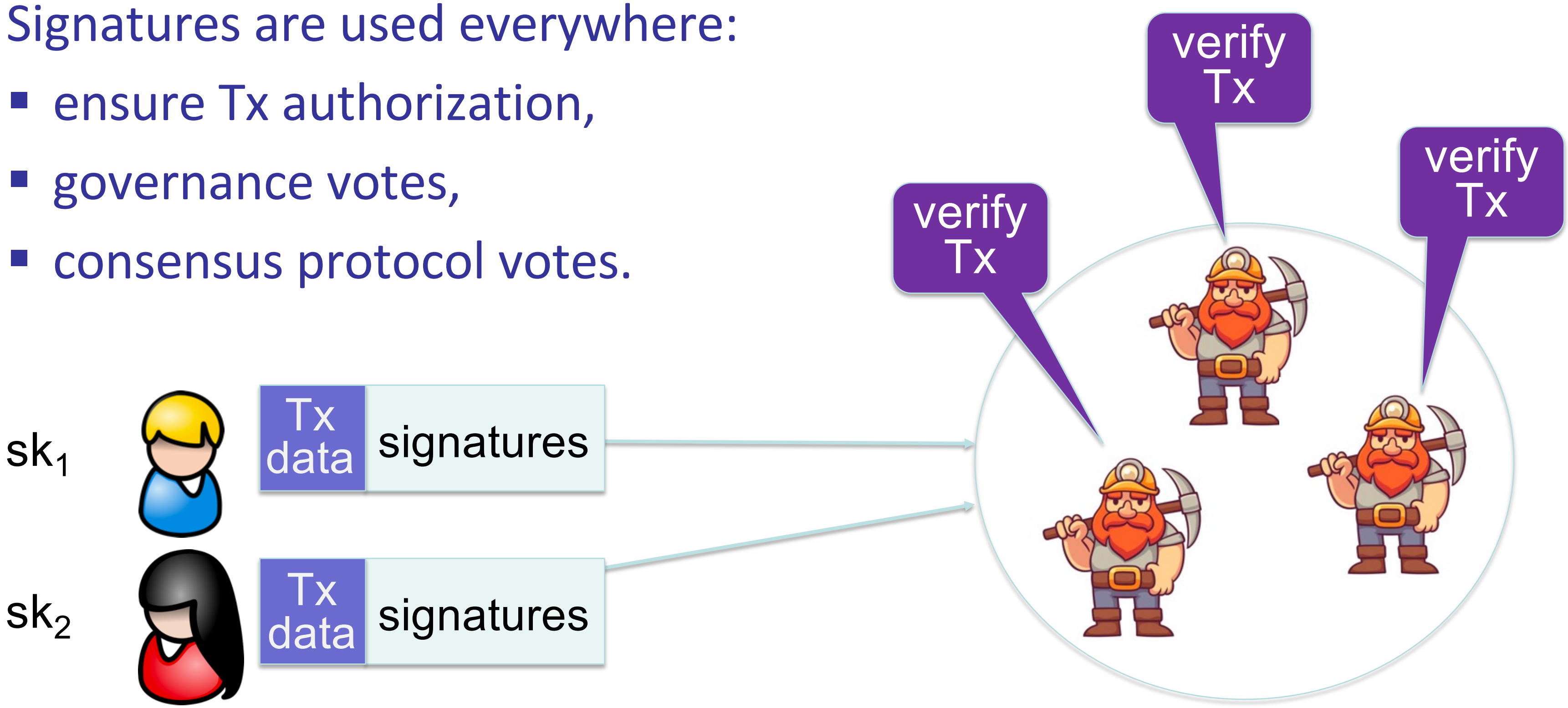
Add Block to the Chain



Verify transactions

Signatures are used everywhere:

- ensure Tx authorization,
- governance votes,
- consensus protocol votes.



Bitcoin

- Pseudonymous: User identities correspond to (*SK*, *PK*) key pairs
- Users digitally sign transactions to authorize movement of money
- Transactions recorded in blockchain

Bitcoin

- Except that the blockchain is *fully decentralized*
- Instead of *one* trusted entity, we rely on *whole community*
- Community maintains fully public blockchain / ledger, so that…
- No bank or jurisdiction can suppress transactions

Any Questions?